THEXRA ENTERPRISE SOLUTION PROPOSAL

Architecture Recommendation Brief

Date: 05/10/2026
Status: Draft

1. CLIENT PROBLEM STATEMENT

Fresh food and dairy products expire on grocery shelves prior to customer purchase, leading to high shrink rates, inventory waste, and substantial margin loss.

2. AI RATIONALE & ANALYSIS

"Augmented reality visual overlays empower store associates to spot nearing-expiry items instantly without tedious manual date checks, preventing fresh stock from expiring unsold. THEXRA View delivers lightweight, hands-free spatial product recognition that seamlessly integrates into routine retail floor workflows."

3. PROPOSED SOLUTION CONCEPT

An augmented reality visual inspection system using THEXRA View that highlights batch expiry status directly over shelf items to guide immediate stock rotation and markdown execution.

IMPLEMENTATION APPROACH:

1. Phase 1: In-store camera testing and ERP inventory data schema integration
2. Phase 2: AR scanning and automated optical character recognition calibration for packaging
3. Phase 3: Pilot deployment across dairy and fresh produce aisles followed by enterprise rollout

DETAILED IMPLEMENTATION PLAN:

1. **Discovery** (1 week):Store floor audit summary • Technical integration blueprint
2. **Design** (1.5 weeks):Floor associate AR interface design • Data sync architecture map
3. **Prototype** (1.5 weeks):Interactive shelf-scanning prototype • Date recognition accuracy validation report
4. **Development** (3 weeks):Core THEXRA View retail application • Inventory ERP connector module
5. **Testing** (1.5 weeks):In-aisle user acceptance sign-off • Device battery and scanning performance report
6. **Deployment** (1.5 weeks):Pilot store deployment • Store staff operational runbook
7. **Support** (Ongoing):Continuous technical support • Monthly waste reduction and audit analytics report

SCOPE OF WORK (SOW)

1. PROJECT SCOPE

Delivery of an AR-based expiry detection and shelf auditing application utilizing THEXRA View for fresh food and dairy retail departments.

2. CORE FEATURES

- Real-time visual expiration date scanning via on-device camera
- Color-coded spatial shelf overlays indicating days to expiry
- Guided First-In First-Out (FIFO) stock rotation visual steps
- Automated markdown trigger alerts sent to retail pricing systems
- Daily aisle audit completion reporting for department supervisors

3. DELIVERABLES

- THEXRA View retail application package
- Inventory database synchronization service
- Store management monitoring console
- Associate training quick-reference cards

4. HARDWARE REQUIREMENTS

- Supported enterprise Android/iOS mobile tablets
- Compatible AR smart glasses headsets

5. SOFTWARE MODULES

- THEXRA View runtime software license
- Store associate mobile client application
- Cloud-based operational analytics portal

6. CONTENT ASSETS

- 3D spatial visual alert indicators
- Standard operating procedure visual cards for stock clerks

7. TRAINING PLAN

Two on-site interactive training workshops for store department leads and shift workers, accompanied by video tutorials.

8. DEPLOYMENT STRATEGY

Phased pilot across target fresh departments in selected pilot locations followed by centralized software push.

9. ONGOING SUPPORT

Business hours SLA technical support, bug fixing, model retraining, and quarterly performance reviews.

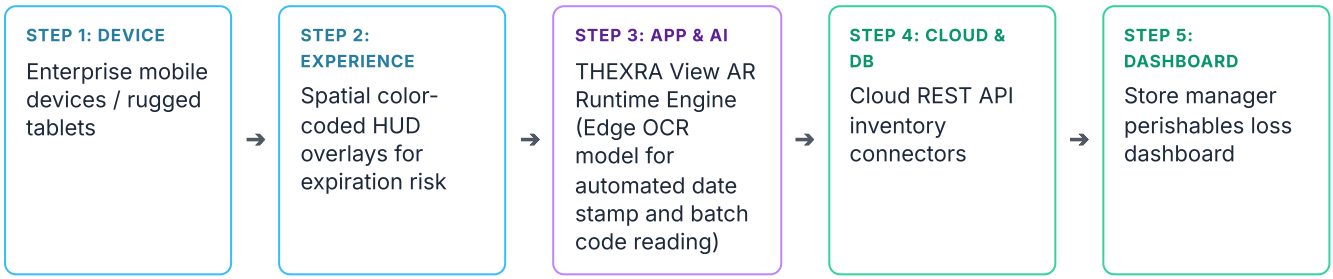
10. EXCLUSIONS (OUT OF SCOPE)

- Procurement of physical store shelves or refrigeration units
- Physical printing hardware for markdown barcode labels
- Deep warehouse logistics beyond the retail store backroom and floor

4. TECHNOLOGY ARCHITECTURE BLUEPRINT

AR — THEXRA View

AUTOMATED SYSTEM FLOW DIAGRAM:



1. Overview: THEXRA View runs on mobile devices or smart glasses, capturing shelf facing data and identifying expiration dates in real time. The engine checks inventory dates against retail ERP thresholds and displays color-coded spatial callouts over products needing rotation or discounting. Audited data synchronizes instantly with central retail store operations dashboards.

2. User Journey: Floor clerk walks the dairy aisle with an AR device pointing at shelves; system highlights nearing-expiry cartons in amber or red; clerk scans and applies dynamic discount labels or moves older stock forward; system logs audit completion automatically.

3. Hardware: Enterprise mobile devices / rugged tablets, RealWear Navigator 520 or lightweight smart glasses, In-store Wi-Fi 6 access points

4. Software: THEXRA View AR Runtime Engine, Retail Inventory Integration Middleware, Dynamic Markdown Pricing Service

5. AI Components: Edge OCR model for automated date stamp and batch code reading, Visual object recognition for planogram compliance

6. XR Components: Spatial color-coded HUD overlays for expiration risk, AR directional indicators for stock rotation sequencing

7. Backend: Cloud REST API inventory connectors, PostgreSQL real-time shelf telemetry database

8. Dashboard: Store manager perishables loss dashboard, Staff audit route and task completion monitor

9. Data Flow: Camera Feed → On-Device Edge OCR → Inventory Backend API → Real-Time AR Spatial Overlay → Store Management Dashboard

Alternative Option: Combination solution — THEXRA Suite (Remote, Make, View)

5. EXPECTED BUSINESS OUTCOMES

35% decrease in perishable product waste, 50% faster daily shelf auditing, and improved gross margins through timely automated markdowns.

6. IMPLEMENTATION NEXT STEPS

Conduct an on-site retail shelf layout audit and test barcode/optical batch-code recognition accuracy under store lighting conditions.